IM-500 Injection Molding Machine — Machine Health & Failure Prediction Manual

For the Failure Prediction Agent. Line: Plastic Molding, Bay 3 | Rev 3.0

1. Monitored Sensors, Thresholds & Failure Signatures

Parameter	Sensor	Normal	Warning	Critical	Failure signature	Predicted lead tin
Hydraulic pump vibration	Vibration (mm/s)	< 2.8	2.8 - 4.5	> 4.5	Rising trend = bearing wear	~7 days to failure
Hydraulic oil temperature	Temp (C)	< 50	50 - 60	> 60	Sustained high = pump/cooling wear	~10 days
Barrel heater temp drift	Temp drift (C)	+/- 5	5 - 10	> 10	Drift up = heater band ageing	~14 days
Main motor current	Current (A)	18 - 24	24 - 28	> 28	Climbing avg = mechanical overload	~5 days
Screw drive torque	Torque (%)	< 70	70 - 85	> 85	Rising trend = screw/barrel wear	~12 days
Clamp tie-bar strain	Strain (uE)	< 800	800 - 1000	> 1000	Imbalance = tie-bar/platen wear	~9 days

2. Failure Prediction Rules

A parameter in the WARNING band for 3 or more consecutive shifts is flagged as a predicted failure. A CRITICAL reading is an immediate predicted failure. The predicted failure date = today + the lead time above. Maintenance must be scheduled BEFORE the predicted failure date.

3. Health Index

Each machine has a Health Index (0-100). Below 60 = schedule maintenance; below 40 = take offline at next changeover. The index drops fastest when vibration and motor current rise together (combined bearing + overload failure).

4. Component-to-Symptom Map (what to inspect)

Symptom	Likely failing component
High vibration + noise	Hydraulic pump bearing
Oil temp high + pressure drop	Hydraulic pump / cooling circuit
Heater temp drift	Barrel heater band + thermocouple
Motor current climbing	Drive motor / mechanical binding
Torque rising	Injection screw / barrel